

RISC-V RV32I Base Instruction Set

[1] RISC-V RV32I Instruction Set Listing, Table 25.2, page 130

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vicilogic RV32I Base Instruction Set

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Interactive RISC-V RV32I Base Instruction Set Table

ID (0x)	ID (0d)	instruction mnemonic	description	instruction(31:0) format							Type	instruction syntax
31121176opcode0												
1	1	lui	load upper immediate	imm[31:12]			rd	0110111		U	lui rd, imm	
2	2	auipc	add upper immed to PC	imm[31:12]			rd	0010111		U	auipc rd, imm	
Jump and link												
3	3	jal	jump and link	imm[20;10;1;11;19:12]			rd	1101111		J	jal rd, imm	
Branch instructions312524201915141211760												
4	4	beq	branch if equal	imm[12;10:5]	rs2	rs1	000	imm[4:1;11]	1100011	B	beq rs1, rs2, imm	
5	5	bne	branch if not equal	imm[12;10:5]	rs2	rs1	001	imm[4:1;11]	1100011	B	bne rs1, rs2, imm	
6	6	blt	branch if less than (lt)	imm[12;10:5]	rs2	rs1	100	imm[4:1;11]	1100011	B	blt rs1, rs2, imm	
7	7	bge	branch if greater than (gt)	imm[12;10:5]	rs2	rs1	101	imm[4:1;11]	1100011	B	bge rs1, rs2, imm	
8	8	bltu	branch if lt than or eq unsgn	imm[12;10:5]	rs2	rs1	110	imm[4:1;11]	1100011	B	bltu rs1, rs2, imm	
9	9	bgeu	branch if gt than unsgn	imm[12;10:5]	rs2	rs1	111	imm[4:1;11]	1100011	B	bgeu rs1, rs2, imm	
Jump and link (registered)31201915141211760												
0xA	10	jalr	jump & link register	imm[11:0]		rs1	000	rd	1100111	I	jalr rd, imm(rs1)	
Memory Load Instructions												
0xB	11	lb	load byte	imm[11:0]		rs1	000	rd	0000011	I	lb rd, imm(rs1)	
0xC	12	lh	load halfword	imm[11:0]		rs1	001	rd	0000011	I	lh rd, imm(rs1)	
0xD	13	lw	load word	imm[11:0]		rs1	010	rd	0000011	I	lw rd, imm(rs1)	
0xE	14	lbu	load byte unsigned	imm[11:0]		rs1	100	rd	0000011	I	lbu rd, imm(rs1)	
0xF	15	lhu	load halfword unsigned	imm[11:0]		rs1	101	rd	0000011	I	lhu rd, imm(rs1)	
Integer register-immediate instructions												
0x10	16	addi	add immediate	imm[11:0]		rs1	000	rd	0010011	I	addi rd, rs1, imm	
0x11	17	slti	set less than immediate	imm[11:0]		rs1	010	rd	0010011	I	slti rd, rs1, imm	
0x12	18	sltiu	set less than immed unsgn	imm[11:0]		rs1	011	rd	0010011	I	sltiu rd, rs1, imm	
0x13	19	xori	xor immediate	imm[11:0]		rs1	100	rd	0010011	I	xori rd, rs1, imm	
0x14	20	ori	or immediate	imm[11:0]		rs1	110	rd	0010011	I	ori rd, rs1, imm	
0x15	21	andi	and immediate	imm[11:0]		rs1	111	rd	0010011	I	andi rd, rs1, imm	
Memory Store Instructions312524201915141211760												
0x16	22	sb	store byte	imm[11:5]		rs2	rs1	000	imm[4:0]	0100011	S	sb rs2, imm(rs1)
0x17	23	sh	store halfword	imm[11:5]		rs2	rs1	001	imm[4:0]	0100011	S	sh rs2, imm(rs1)
0x18	24	sw	store word	imm[11:5]		rs2	rs1	010	imm[4:0]	0100011	S	sw rs2, imm(rs1)
Constant shift Instructionsf7												
0x19	25	slli	constant-shift left	0000000		shamt	rs1	001	rd	0010011	I	slli rd, rs1, shamt
0x1A	26	srl	constant-shift right	0000000		shamt	rs1	101	rd	0010011	I	srl rd, rs1, shamt
0x1B	27	srai	constant-shift right arithmetic	0100000		shamt	rs1	101	rd	0010011	I	srai rd, rs1, shamt
Integer register-register instructions												
0x1C	28	add	add	0000000		rs2	rs1	000	rd	0110011	R	add rd, rs1, rs2
0x1D	29	sub	subtract	0100000		rs2	rs1	000	rd	0110011	R	sub rd, rs1, rs2
0x1E	30	sll	register-shift left	0000000		rs2	rs1	001	rd	0110011	R	sll rd, rs1, rs2
0x1F	31	slt	set less than	0000000		rs2	rs1	010	rd	0110011	R	slt rd, rs1, rs2
0x20	32	sltu	set less than unsigned	0000000		rs2	rs1	011	rd	0110011	R	sltu rd, rs1, rs2
0x21	33	xor	xor	0000000		rs2	rs1	100	rd	0110011	R	xor rd, rs1, rs2
0x22	34	srl	register-shift right (logical)	0000000		rs2	rs1	101	rd	0110011	R	srl rd, rs1, rs2
0x23	35	sra	register-shift right (arithmetic)	0100000		rs2	rs1	101	rd	0110011	R	sra rd, rs1, rs2
0x24	36	or	or	0000000		rs2	rs1	110	rd	0110011	R	or rd, rs1, rs2
0x25	37	and	and	0000000		rs2	rs1	111	rd	0110011	R	and rd, rs1, rs2
31201915141211760												
0x26	38	FENCE		fm	pred	succ	rs1	000	rd	0001111		
0x27	39	ECALL		000000000000			00000	000	00000	1110011		
0x28	40	EBREAK		000000000001			00000	000	00000	1110011		